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STEP-DOWN PRINTED CIRCUIT BOARD WITH A FLEXIBLE CONNECTOR

Abstract

A semiconductor package includes a step-down PCB. The step-down PCB includes a first portion having a first height a second portion having a second height that is less than the first height. The step-down PCB can be a single, unitary structure or have different PCB portions. When the step-down PCB has different PCB portions, a flexible connector mechanically and electrically couples the PCB portions together. Due to the reduced thickness of the second portion, additional semiconductor dies can be added to a stack of semiconductor dies, thereby increasing the capabilities of the semiconductor package. Additionally, the first portion includes contacts for a connector. The connector is used to communicatively couple the semiconductor package to an associated connector slot of a host device.

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Background/Summary

BACKGROUND

[0001] Semiconductor packages, such as non-volatile memory devices, are widely used in various electronic devices such as cellular telephones, digital cameras, personal digital assistants, medical electronics, mobile computing devices, and non-mobile computing devices. In examples in which the semiconductor package is a non-volatile memory device, the semiconductor package includes a stack of memory dies, such as NAND memory dies. The capacity of the non-volatile memory device can be increased by adding additional memory dies to the stack. However, as additional memory dies are added to the stack, the height of the semiconductor package increases.

[0002] Increasing the height of the semiconductor package may not be feasible for a number of reasons. For example, electronic devices continue to get thinner and a semiconductor package having an increased height may not fit within the housing of the electronic device. In another example, the total height of the semiconductor package may be specified by one or more industry standards and/or constraints.

[0003] Accordingly, it would be beneficial for a semiconductor package to have increased capacity and/or capabilities without increasing the height of the semiconductor package.

SUMMARY

[0004] The present application describes a semiconductor package having a tiered or a step-down printed circuit board (PCB). The tiered or step-down PCB includes a first portion having a first thickness or height and a second portion having a second thickness or height that is less than the first thickness or height. In an example, the PCB is a single, unitary structure. In another example, the first portion and the second portion are separate PCBs but are connected by a flexible connector (e.g., a rigid flex).

[0005] In an example, various semiconductor dies and/or passive components are mounted or are otherwise coupled to the thinner portion of the PCB. As such, more semiconductor dies can be provided on the second portion when compared with solutions in which different portions of the PCB does not have different heights. For example, the increased height of a memory die stack can be offset by the reduced thickness of the PCB.

[0006] The first portion of the PCB, or the thicker portion of the PCB, includes contacts (e.g., gold finger contacts or other types of contacts) for a connector. The connector is used to communicatively couple the semiconductor package to an associated connector slot of a host device. In an example, the connector is a M.2 connector. In another example, the connector is an Enterprise and Datacenter Standard Form Factor (EDSFF) connector.

[0007] Accordingly, the present application describes a semiconductor package that includes a PCB portion having a first thickness and a second PCB portion having a second thickness that is less than the first thickness. A flexible connector is provided between the first PCB portion and the second PCB portion. In an example, the flexible connector mechanically and electrically connects the first PCB portion and the second PCB portion. A plurality of contacts are provided on the first PCB portion for electrically coupling the semiconductor package to a connector slot of a host device. Additionally, a plurality of semiconductor dies are provided on the second PCB portion.

[0008] Other examples describe a semiconductor package that includes a plurality of contacts provided on a first PCB portion. In an example, the first PCB portion also includes a first number of layers and has a first thickness. A plurality of semiconductor dies are provided on a second

[0009] PCB portion. In an example, the second PCB portion is separate from the first PCB portion and has a second number of layers that is less than the first number of layers. The second PCB portion also has a second thickness that is less than the first thickness. In an example, a flexible connector couples the first PCB portion and the second PCB portion.

[0010] The present application also describes a semiconductor package that includes a first PCB portion having a first thickness and a second PCB portion having a second thickness that is less than the first thickness. A flexible connection means is provided between the first PCB portion and the second PCB portion. In an example, the flexible connection means mechanically and electrically couples the first PCB portion and the second PCB portion. The semiconductor package also includes a plurality of signal carrying means provided on a surface of the first PCB portion. Additionally, a plurality of semiconductor dies is provided on the second PCB portion.

[0011] This summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used to limit the scope of the claimed subject matter.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0012] Non-limiting and non-exhaustive examples are described with reference to the following Figures.

[0013] FIG. 1 illustrates a step-down printed circuit board (PCB) according to an example.

[0014] FIG. 2A illustrates a step-down PCB having a first PCB portion and a second PCB portion connected by a flex connector according to an example.

[0015] FIG. 2B illustrates a step-down PCB having a first PCB portion and a second PCB portion connected by a flexible connector according to another example.

[0016] FIG. 3A illustrates a semiconductor package having a step-down PCB according to an example.

[0017] FIG. 3B illustrates a top view of the semiconductor package of FIG. 3A according to an example.

[0018] FIG. 4A illustrates a semiconductor package in which one of the PCB portions is encapsulated according to an example.

[0019] FIG. 4B illustrates a semiconductor package in which a PCB portion is encapsulated and includes a heat dissipation feature according to an example.

[0020] FIG. 4C illustrates a semiconductor package in which a PCB portion is encapsulated and includes multiple heat dissipation features according to an example.

DETAILED DESCRIPTION

[0021] In the following detailed description, references are made to the accompanying drawings that form a part hereof, and in which are shown by way of illustrations specific embodiments or examples. These aspects may be combined, other aspects may be utilized, and structural changes may be made without departing from the present disclosure. The following detailed description is therefore not to be taken in a limiting sense, and the scope of the present disclosure is defined by the appended claims and their equivalents.

[0022] The demand for semiconductor packages, such as non-volatile memory devices, is increasing. However, as the demand increases, so do the demands for higher capacity and better performance. In examples in which the semiconductor package is a non-volatile memory device, such as a NAND memory device, the non-volatile memory device includes a number of NAND memory dies stacked on top of one another. To increase capacity, additional memory dies are added to the stack.

[0023] As additional memory dies are added to the stack, the height of the semiconductor package increases. However, it may not be feasible to increase the height of the semiconductor package. For example, as electronic devices continue to get thinner, a thicker/taller semiconductor package may not fit within the housing of the electronic device. In another example, a height or thickness of the semiconductor package may need to comply with industry standards or constraints.

[0024] To address the above, the present application describes a semiconductor package that has a tiered or a step-down printed circuit board (PCB). In an example, the step-down PCB includes a first portion having a first thickness or height and a second portion having a second thickness or height. In an example, the second thickness or height is less than the first thickness or height.

[0025] The step-down PCB of the present disclosure has a number of different configurations. In one configuration, the step-down PCB is a single, unitary structure. In another configuration, the first portion and the second portion are separate PCBs. However, in this configuration, the first PCB portion and the second PCB portion are electrically and mechanically coupled using a flexible connector such as, for example, a rigid flex.

[0026] The first portion, or the thicker portion, of the PCB includes contacts (e.g., gold finger contacts or other types of contacts) for a connector. The connector is used to communicatively couple the semiconductor package to an associated connector slot of a host device. In an example, the connector is a M.2 connector. In another example, the connector is an Enterprise and Datacenter Standard Form Factor (EDSFF) connector.

[0027] In some examples, the first portion has a thickness or height that matches the thickness or height of the standard connector. For example, if the height of a standard M.2 connector is 0.8 millimeters (mm), the thickness or the height of the first portion is 0.8 mm.

[0028] In an example, various semiconductor dies and/or passive components are mounted (e.g., surface mounted) on or are otherwise coupled to the second/thinner portion of the PCB. As such, more semiconductor dies can be provided on the second portion when compared with current solutions. For example, the increased height of a memory die stack due to an increased number of semiconductor dies is offset by the reduced thickness of the second portion.

[0029] Accordingly, many technical benefits may be realized including, but not limited to, increasing the capacity and/or capabilities of a semiconductor package without increasing a height or thickness of the semiconductor package; enabling the semiconductor package to utilize standard connectors without requiring a host device to have a new type of connector; and allowing flexibility of placement and/or packaging options due to the use of a flexible connector.

[0030] These benefits, along with other examples, will be shown and described in greater detail with respect to FIG. 1-FIG. 4C.

[0031] FIG. 1 illustrates a step-down printed circuit board (PCB) **100** according to an example. Although a PCB is specifically mentioned, the examples described herein are applicable to a substrate or to any other material on which electronic components are placed and/or mounted.

[0032] In an example, the step-down PCB **100** is a single, unitary piece. For example, the step-down PCB **100** includes a first portion **110** having a first thickness or height (designated by “H”) and a second portion **120** having a second thickness or height (designated by “h”). In an example, the first portion **110** of the step-down PCB **100** has the first thickness to enable the first portion **110** to act as a connector. For example and as will be described in greater detail below, the first portion **110** includes a number of contacts that enables the first portion **110** to be received into a standard connector slot of a host device.

[0033] In an example, the height H (or the thickness) of the first portion **110** of the step-down PCB is 0.8 millimeters (mm). In such an example, the first portion **110**, along with the various contacts that are provided on the first portion, are used as an M.2 connector. In another example, the height H of the first portion **110** of the step-down PCB **100** is 1.57 mm. In such an example, the first portion **110**, along with the various contacts provided on the first portion **110**, are used as an EDSFF connector. Although an M.2 connector and an EDSFF connector are specifically

mentioned, the height and/or thickness of the first portion **110** may be any height or thickness depending on the desired type of connector.

[0034] In an example, the thickness or height h of the second portion **120** may also vary. For example, when the step-down PCB **100** is used as part of a non-volatile memory device, the height h of the second portion **120** is dependent on a desired capacity of the non-volatile memory device and/or on the number of memory dies in the stack. In another example, the height h of the second portion **120** is dependent on a type of cover that will be placed on or over the various semiconductor dies and/or passive components placed on/in the second portion **120**.

[0035] In yet another example, the height h of the second portion **120** is dependent on the type of semiconductor package and/or on the size of the first portion **110**. For example, if the first portion **110** of the step-down PCB **100** has a height H of 1.57 mm, the second portion **120** has a height h of 0.95 mm. In another example, if the first portion **110** of the step-down PCB **100** has a height H of 0.8 mm, the second portion **120** has a height h of 0.2 mm. Although specific dimensions are given, these are for example purposes only.

[0036] In the example shown, the step-down PCB **100** is fabricated such that the second portion **120** extends from a lower or bottom portion of the first portion **110**. For example, a bottom surface of the second portion **120** is planar to a bottom surface of the first portion **110**. However, in other examples, the second portion **120** extends from any portion (e.g., a top portion, a middle portion) of the first portion **110**. In an example, during fabrication, one or more layers of the second portion **120** are removed which causes the second portion **120** to have the height h .

[0037] The first portion **110** and the second portion **120** may have any length. In an example, the length of the first portion **110** is dependent on the number and/or type of contacts that are used to make up the connector. In another example, the length of the first portion **110** is dependent on an arrangement of traces or other communication lines that are used to connect the contacts on the first portion **110** with the semiconductor dies on the second portion **120**. In an example, the length of the second portion **120** is dependent on the number or type of semiconductor dies that will be placed on the second portion **120** and/or the number or type of passive components that will be placed on the second portion **120**.

[0038] FIG. 2A illustrates a step-down PCB **200** having a first PCB portion **210** and a second PCB portion **220** connected by a flexible connector **230** according to an example. In an example, the flexible connector **230** is a rigid flex. Although a rigid flex is specifically mentioned, the first portion **210** and the second portion **220** are connectable by any flexible and/or rigid material that provides both an electrical connection and a mechanical connection between each portion and/or between the various components on each portion.

[0039] In an example, the first portion **210** has a first thickness or height (designated by “ H ”). Likewise, the second portion **220** has a second thickness or height (designated by “ h ”). Like the example shown in FIG. 1, the first thickness of the first portion **210** enables the first portion **210** to act as a connector.

[0040] Additionally, in some examples, the height H (or the thickness) of the first portion **210** of the step-down PCB **200** is 0.8 millimeters (mm). In another example, the height H of the first portion **210** of the step-down PCB **200** is 1.57 mm. Although specific thicknesses or heights are mentioned, the height and/or thickness of the first portion **210** may be any height or thickness depending on a type of connector that is needed.

[0041] In an example, the thickness or height h of the second portion **220** also varies. For example, when the step-down PCB **200** is used as part of a non-volatile memory device, the height h of the second portion **220** is dependent on a desired capacity of the non-volatile memory device and/or on the number of memory dies in the stack. In another example, the height h of the second portion **220** is dependent on a type of cover that will be placed on or over the various semiconductor dies and/or passive components placed on/in the second portion **220**.

[0042] In an example, the flexible connector **230** mechanically and/or electrically couples the first

portion **210** of the step-down PCB **200** to the second portion **220** of the step-down PCB **200**. In an example, the flexible connector **230** enables the first portion **210** and/or the second portion **220** to move or pivot relative to one another (e.g., in the direction of arrows **240**). In another example, the flexible connector enables the first portion **210** and/or the second portion **220** to move laterally with respect to one another (e.g., in the direction of arrows **245**). As a result, the overall length of the step-down PCB **200** may be adjusted (e.g., shortened).

[0043] The flexible connector **230** mechanically and/or electrically couples to portion by any suitable means. For example, a proximal end of the flexible connector **230** (represented by the dashed lines) is coupled to and/or sandwiched between one or more layers of the first portion **210** of the step-down PCB **200**. Likewise, a distal end of the flexible connector **230** is coupled to and/or sandwiched between one or more layers of the second portion **220** of the step-down PCB **200**. In an example, the flexible connector **230** includes traces, signal lines, or other communication paths that enable the semiconductor dies and/or passive components on the second portion **220** to be electrically and/or communicatively coupled to the respective contacts on the first portion **210**.

[0044] In an example, a protective and/or flexible coating or cover **235** is applied to the flexible connector **230**. The flexible cover **235** may be made from any material and is used to protect and/or insulate the traces, signal lines or other communication paths that extend through the flexible connector **230** and communicatively couple the first portion **210** to the second portion **220**.

[0045] In an example, the flexible connector **230** is coupled to a bottom portion of the first portion **210** and a bottom portion of the second portion **220**. For example, the flexible connector **230** is coupled to each portion such that a bottom surface of the second portion **220** is planar to a bottom surface of the first portion **210**. However, in other examples, the flexible connector **230** is coupled to any portion of the first portion **210** and/or the second portion **220**.

[0046] For example and referring to FIG. 2B, FIG. 2B illustrates a step-down PCB **250** having a first PCB portion **260** and a second PCB portion **270** connected by a flexible connector **280** according to another example. In an example, the step-down PCB **250** is similar to the step-down PCB **200** shown and described with respect to FIG. 2A. However, in this example, the flexible connector **280** extends from, or is otherwise coupled to a middle portion of the first portion **260** and to a middle portion of the second portion **270**.

[0047] In the examples shown in FIG. 2A and FIG. 2B, each portion may have any length. For example, the length of the first portion **210** of the step-down PCB **200** is dependent on the number and/or type of contacts that are used to make up the connector. In another example, the length of the first portion **210** is dependent on an arrangement of traces or other communication lines that are used to connect the contacts with the semiconductor dies on the second portion **220** of the step-down PCB **200**. In an example, the length of the second portion **220** is dependent on the number or type of semiconductor dies that will be placed on the second portion **220** and/or the number or type of passive components that will be placed on the second portion **220**.

[0048] FIG. 3A illustrates a semiconductor package **300** having a step-down PCB according to an example. In the example shown in FIG. 3A, the step-down PCB is similar to the step-down PCB **200** shown and described with respect to FIG. 2A. However, the semiconductor package **300** may have any of the various step-down PCBs shown and described herein.

[0049] In this example, the step-down PCB includes a first portion **310** and a second portion **320**. The second portion **320** is separate, and apart from, the first portion **310**. In an example, the first portion **310** includes a first number of layers **380** and the second portion **320** includes a second number of layers **385**.

[0050] In an example, and although the first portion **310** and the second portion **320** are separate and apart from one another, a flexible connector **330** electrically and/or mechanically couples the first portion **310** and the second portion **320**. In an example, the flexible connector **330** is a rigid flex, although any connector may be used to communicatively couple the first portion **310** to the second portion **320**.

[0051] In an example, the first portion **310** includes a number of contacts **340** (e.g., gold finger contacts) that comprise a connector. In an example, the connector is a M.2 connector. In another example, the connector is an EDSFF connector. Although specific connectors are described, the contacts **340** may be part of any desired connector type. Likewise, a height or thickness of the first portion **310** may be based, at least in part, on the type of connector desired.

[0052] For example, if the connector is a M.2 connector, the height or thickness of the first portion **310** is 0.8 mm. However, if the connector is an EDSFF connector, the height or thickness of the first portion **310** is 1.57 mm. Although specific heights are described, the first portion **310** may have any height based on a desired connector type and/or on the dimensions of a connector slot of a host device to which the semiconductor package **300** will be communicatively coupled.

[0053] As also shown in FIG. **3A**, the second portion **320** includes a number of semiconductor dies and/or passive components. In an example, the semiconductor dies and/or the passive components are surface mounted or are otherwise electrically coupled to the second portion **320**. For example, the semiconductor package **300** includes a number of memory dies **350** arranged in a stacked configuration. In this example, each of the memory dies **350** in the stack is communicatively coupled to the second portion **320** using one or more bond wires **355**.

[0054] In an example, an application specific integrated circuit (ASIC) **360** is also mounted on, or is otherwise associated with, the second portion **320**. In an example, the ASIC **360** is a controller or a processor. Although a single ASIC **360** is shown, multiple ASICs **360** may be provided on, or associated with, the second portion **320**.

[0055] In addition to the semiconductor dies, the semiconductor package **300** also includes passive components **370**. In an example, the passive components **370** are also mounted to the second portion **320** of the step-down PCB. Although specific components and semiconductor dies are mentioned, the semiconductor package **300** may include any number of different electronic components, semiconductor devices and/or dies.

[0056] FIG. **3B** illustrates a top view of the semiconductor package **300** of FIG. **3A** according to an example. As shown in FIG. **3B**, the semiconductor package **300** includes the first portion **310** and the second portion **320**. The flexible connector **330** mechanically and electrically couples the first portion **310** and the second portion **320** such as previously described.

[0057] In an example, the first portion **310** includes a plurality of contacts **340** provided on an upper surface. However, in an example, additional contacts are provided on a bottom surface of the first portion **310**.

[0058] In an example, signal lines **390** electrically and/or communicatively couple the contacts **340** to respective semiconductor dies on the second portion **320**. For example, one or more signal lines **390** extend through the first portion **310**, through the flexible connector **330**, and to respective bond pads **365** associated with the stack of semiconductor dies **350**. In another example, one or more signal lines **390** extend through the first portion **310**, through the flexible connector **330**, and to the ASIC **360**. Additional signal lines **390** may also electrically couple the passive components **370** to one or more of the various semiconductor dies and/or the contacts **340**.

[0059] FIG. **4A** illustrates a semiconductor package **400** in which one of the PCB portions is encapsulated according to an example. In an example, the semiconductor package **400** is similar to the semiconductor package **300** shown and described with respect to FIG. **3A** and FIG. **3B**. For example, the semiconductor package **400** includes a number of different semiconductor dies and passive components. Additionally, the semiconductor package **400** includes a step-down PCB having a first portion **410**, a second portion **420** separate from the first portion **410** and a flexible connector **430**. However, the semiconductor package **400** may have any of the various step-down PCBs shown and described herein.

[0060] In this example, the second portion **420** includes a cover **440**. In an example, the cover **440** is a mechanical cover. In another example, the cover **440** is a molding compound. In yet another example, the cover **440** is a mechanical cover than surrounds or otherwise encapsulates a molding

compound. However, in this example, the cover **440** only encapsulates the second portion **420** (or at least some of the passive components and/or semiconductor dies on the second portion **420**) but not the flexible connector **430** or the first portion **410**.

[0061] FIG. **4B** illustrates a semiconductor package **400** in which a PCB portion is encapsulated and includes a heat dissipation **450** feature according to an example. Like the example shown and described with respect to FIG. **4A**, the semiconductor package includes a step-down PCB having a first portion **410**, a second portion **420** separate from the first portion **410** and a flexible connector **430**.

[0062] In this example, the second portion **420** includes a heat dissipation feature **450**. In an example, the heat dissipation feature **450** is part of, or is otherwise included with, a cover (e.g., cover **440** (FIG. **4A**) or other material that encapsulates the various components on the second portion **420**. In an example, the heat dissipation feature **450** is a silicon layer. Although a heat dissipation feature **450** is specifically mentioned, an electromagnetic interference (EMI) layer may be included in lieu of, or in combination with, the heat dissipation feature **450**.

[0063] In an example, the heat dissipation feature **450** contacts a top surface of a semiconductor die stack and/or an ASIC. In some examples, the heat dissipation feature **450** extends across the entire length of the cover and/or the second portion **420**. In another example, the heat dissipation feature **450** is broken up into different segments or portions. In such an example, a first portion of the heat dissipation feature **450** contacts a top surface of the semiconductor die stack while a second portion of the heat dissipation **450** feature contacts a top surface of the ASIC. The heat dissipation feature **450** conducts heat away from the ASIC and/or the stack of semiconductor dies and into the cover or other material that surrounds or encapsulates the second portion **420**.

[0064] FIG. **4C** illustrates a semiconductor package **400** in which a PCB portion is encapsulated and includes multiple heat dissipation features according to an example. Like the examples shown in FIG. **4A** and FIG. **4B**, the semiconductor package **400** includes a step-down PCB having a first portion **410**, a second portion **420** separate from the first portion **410** and a flexible connector **430**.

[0065] In this example, the second portion **420** includes multiple heat dissipation features. For example, the second portion **420** includes the heat dissipation feature **450** shown and described with respect to FIG. **4B**. Additionally, the second portion **420** includes a chimney or trench **460** that dissipates heat from within the second portion **420** of the semiconductor package **400** and to the heat dissipation feature **450**. In an example, the trench **460** is filled with a thermally conductive material, such as silver, although other materials may be used. Additionally, although a single trench **460** is shown, multiple trenches **460** may be provided in the second portion **420** of the semiconductor package **400**.

[0066] Based on the above, examples of the present disclosure describe a semiconductor package, comprising: a first printed circuit board (PCB) portion having a first thickness; a second PCB portion having a second thickness that is less than the first thickness; a flexible connector provided between the first PCB portion and the second PCB portion, the flexible connector mechanically and electrically connecting the first PCB portion and the second PCB portion; a plurality of contacts provided on the first PCB portion for electrically coupling the semiconductor package to a connector of a host device; and a plurality of semiconductor dies provided on the second PCB portion. In an example, the semiconductor package also includes one or more passive components mounted on the second PCB portion. In an example, at least one of the plurality of semiconductor dies is a NAND memory die. In an example, at least one of the plurality of semiconductor dies is an application-specific integrated circuit (ASIC). In an example, the semiconductor package also includes a molding compound at least partially covering the second PCB portion. In an example, the plurality of contacts are part of an M.2 connector. In an example, the plurality of contacts are part of an Enterprise and Datacenter Standard Form Factor (EDSFF) connector. In an example, the flexible connector extends from a top portion of the first PCB portion. In an example, the flexible connector extends from a middle portion of the first PCB portion. In an example, the flexible

connector extends from a top portion of the second PCB portion. In an example, the flexible connector extends from a middle portion of the second PCB portion.

[0067] Examples also describe a semiconductor package, comprising: a plurality of contacts provided on a first printed circuit board (PCB) portion, the first PCB portion having a first number of layers and a first thickness; a plurality of semiconductor dies provided on a second PCB portion, the second PCB portion being separate from the first PCB portion and having a second number of layers that is less than the first number of layers and having a second thickness that is less than the first thickness; and a flexible connector coupling the first PCB portion and the second PCB portion. In an example, the semiconductor package also includes a cover at least partially encapsulating the second PCB portion. In an example, the plurality of contacts form an M.2 connector. In an example, the plurality of contacts form an Enterprise and Datacenter Standard Form Factor (EDSFF) connector. In an example, the flexible connector extends from a top portion of the first PCB portion. In an example, the flexible connector extends from a middle portion of the second PCB portion.

[0068] Additional examples describe a semiconductor package, comprising: a first printed circuit board (PCB) portion having a first thickness; a second PCB portion having a second thickness that is less than the first thickness; a flexible connection means provided between the first PCB portion and the second PCB portion, the flexible connection means mechanically and electrically coupling the first PCB portion and the second PCB portion; a plurality of signal carrying means provided on a surface of the first PCB portion; and a plurality of semiconductor dies provided on the second PCB portion. In an example, the semiconductor package also includes a covering means provided on the second PCB portion. In an example, the semiconductor dies are NAND memory dies.

[0069] The description and illustration of one or more aspects provided in the present disclosure are not intended to limit or restrict the scope of the disclosure in any way. The aspects, examples, and details provided in this disclosure are considered sufficient to convey possession and enable others to make and use the best mode of claimed disclosure.

[0070] The claimed disclosure should not be construed as being limited to any aspect, example, or detail provided in this disclosure. Regardless of whether shown and described in combination or separately, the various features are intended to be selectively rearranged, included or omitted to produce various embodiments with a particular set of features. Having been provided with the description and illustration of the present application, one skilled in the art may envision variations, modifications, and alternate aspects falling within the spirit of the broader aspects of the general inventive concept embodied in this application that do not depart from the broader scope of the claimed disclosure.

[0071] References to an element herein using a designation such as “first,” “second,” and so forth does not generally limit the quantity or order of those elements. Rather, these designations may be used as a method of distinguishing between two or more elements or instances of an element. Thus, reference to first and second elements does not mean that only two elements may be used or that the first element precedes the second element. Additionally, unless otherwise stated, a set of elements may include one or more elements.

[0072] Terminology in the form of “at least one of A, B, or C” or “A, B, C, or any combination thereof” used in the description or the claims means “A or B or C or any combination of these elements.” For example, this terminology may include A, or B, or C, or A and B, or A and C, or A and B and C, or 2A, or 2B, or 2C, or 2A and B, and so on. As an additional example, “at least one of: A, B, or C” is intended to cover A, B, C, A-B, A-C, B-C, and A-B-C, as well as multiples of the same members. Likewise, “at least one of: A, B, and C” is intended to cover A, B, C, A-B, A-C, B-C, and A-B-C, as well as multiples of the same members.

[0073] Similarly, as used herein, a phrase referring to a list of items linked with “and/or” refers to any combination of the items. As an example, “A and/or B” is intended to cover A alone, B alone,

or A and B together. As another example, “A, B and/or C” is intended to cover A alone, B alone, C alone, A and B together, A and C together, B and C together, or A, B, and C together.

Claims

1. A semiconductor package, comprising: a first printed circuit board (PCB) portion having a first thickness; a second PCB portion having a second thickness that is less than the first thickness; a flexible connector provided between the first PCB portion and the second PCB portion, the flexible connector mechanically and electrically connecting the first PCB portion and the second PCB portion; a plurality of contacts provided on the first PCB portion for electrically coupling the semiconductor package to a connector of a host device; and a plurality of semiconductor dies provided on the second PCB portion.
2. The semiconductor package of claim 1, further comprising one or more passive components mounted on the second PCB portion.
3. The semiconductor package of claim 1, wherein at least one of the plurality of semiconductor dies is a NAND memory die.
4. The semiconductor package of claim 1, wherein at least one of the plurality of semiconductor dies is an application-specific integrated circuit (ASIC).
5. The semiconductor package of claim 1, further comprising a molding compound at least partially covering the second PCB portion.
6. The semiconductor package of claim 1, wherein the plurality of contacts are part of an M.2 connector.
7. The semiconductor package of claim 1, wherein the plurality of contacts are part of an Enterprise and Datacenter Standard Form Factor (EDSFF) connector.
8. The semiconductor package of claim 1, wherein the flexible connector extends from a top portion of the first PCB portion.
9. The semiconductor package of claim 1, wherein the flexible connector extends from a middle portion of the first PCB portion.
10. The semiconductor package of claim 1, wherein the flexible connector extends from a top portion of the second PCB portion.
11. The semiconductor package of claim 1, wherein the flexible connector extends from a middle portion of the second PCB portion.
12. A semiconductor package, comprising: a plurality of contacts provided on a first printed circuit board (PCB) portion, the first PCB portion having a first number of layers and a first thickness; a plurality of semiconductor dies provided on a second PCB portion, the second PCB portion being separate from the first PCB portion and having a second number of layers that is less than the first number of layers and having a second thickness that is less than the first thickness; and a flexible connector coupling the first PCB portion and the second PCB portion.
13. The semiconductor package of claim 12, further comprising a cover at least partially encapsulating the second PCB portion.
14. The semiconductor package of claim 12, wherein the plurality of contacts form an M.2 connector.
15. The semiconductor package of claim 12, wherein the plurality of contacts form an Enterprise and Datacenter Standard Form Factor (EDSFF) connector.
16. The semiconductor package of claim 12, wherein the flexible connector extends from a top portion of the first PCB portion.
17. The semiconductor package of claim 12, wherein the flexible connector extends from a middle portion of the second PCB portion.
18. A semiconductor package, comprising: a first printed circuit board (PCB) portion having a first thickness; a second PCB portion having a second thickness that is less than the first thickness; a

flexible connection means provided between the first PCB portion and the second PCB portion, the flexible connection means mechanically and electrically coupling the first PCB portion and the second PCB portion; a plurality of signal carrying means provided on a surface of the first PCB portion; and a plurality of semiconductor dies provided on the second PCB portion.

19. The semiconductor package of claim 18, further comprising a covering means provided on the second PCB portion.

20. The semiconductor package of claim 18, wherein the semiconductor dies are NAND memory dies.
